

ANUBHAV SAINI

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TECHNICAL SKILLS

PCB Design: Altium Designer, PCB & Flex Layout, Schematic Capture, High-Voltage Design (HIPOT, IPC Class 3), DFM
Power & Validation: Power Electronics (DC-DC), Signal Integrity, SPICE, Oscilloscopes & Probing, Soldering & Rework
Embedded & Software: Python, C/C++, STM32, SPI, I2C, UART, CAN, USB, Ethernet, MATLAB, OpenCV, ROS, Linux

EDUCATION

Bachelor of Applied Science in Engineering Physics **Expected May 2027**
University of British Columbia – Specialization in Electrical Engineering *Vancouver, BC*

EXPERIENCE

Electronics Hardware Engineering Intern **January 2026 – August 2026**
Kardium Inc. *Burnaby, BC*

- Designed 6-layer PCB for Kardium's **highest-volume medical device (10,000+ units/year)** to increase HIPOT safety margin and reduce field-failure risk by re-architecting High Voltage/Low Voltage routing and optimizing the isolation stack-up.
- Re-defined PCB design rules and updated fabrication/assembly documentation during a vendor transition to **lower production costs and ensure manufacturability** under new process constraints by analyzing DFM limitations, capabilities and rules.
- Created flex PCB layouts carrying 1500V+ signals to enable **short-term cost reduction** without compromising creepage/clearance compliance by applying HV spacing rules, optimizing bend-area routing, and validating against IPC Class 3 rules.
- Led characterization and validation of a 12-layer RF Ablation PCB to **achieve 100% test coverage** for design verification of a safety-critical system by authoring and executing 25+ tests across Power, SI, ADC, Isolation boundary, and Thermal suites.
- Debugged multiple 14-layer boards within a 10+ board integrated system to resolve system-level failures and accelerate hardware bring-up by performing high-speed signal analysis, targeted probing, and precision rework to isolate cross-board interactions.

Hardware Engineering Intern **May 2025 – December 2025**
Arlo Technologies *Vancouver, BC*

- Designed a Power ORing Switch with **99% efficiency** to ensure the camera always charged from the strongest available source by using transistor and diode logic to build an analog control system with integrated FW control and fallback HW logic.
- Led R&D of a solar-powered** security camera by defining **system architecture** (multiple power inputs configured with Buck-Boost/charger/fuel gauge) and **validating design** through power, efficiency, and battery charge cycling.
- Reduced sensor system **yield loss by 30%** to restore accurate motion detection across all operating states and prevent false positive readings by resolving PIR noise caused by RF radiation through hardware-level debugging and fixes.
- Increased test coverage for PIR and ambient light sensors from 60% to **full coverage** to ensure complete validation in wireless environments by creating a test suite that captured all operating states and RF related edge cases.

Electrical Engineering Intern **September 2024 – December 2024**
Sarcomere Dynamics Inc. *Vancouver, BC*

- Designed a **Motor Driver PCB in Altium** – SPI to the BLDC motor driver, I2C magnetometer feedback for rotor positioning, CAN for multi-board scalability, and **3.3V** on-board linear regulation all within a heavily size-constrained board.
- Developed python firmware for an in-house fingertip force sensor to create a **low-cost alternative** to OTS solutions, enabling real-time reconstruction of normal force and shear direction **with 90% accuracy** through tri-axis magnetometer processing.
- Created a lightweight and simple calibration method to **improve sensor reliability**, using pseudo-inverse matrix mapping and validating performance through K-Means clustering to ensure consistent and accurate force interpretation.

Product Engineering Intern **May 2024 – August 2024**
Microchip Technology *Burnaby, BC*

- Programmed Python test scripts and custom firmware to characterize the SERDES I3C Pad, executing **8 AC and 5 DC tests** across **5 voltage/temperature (VT) corners** with per-test BERT/DCSU configuration and pass/fail yield checks.
- Developed a multi-threaded testing procedure to reduce characterization time by **more than 40%**, holding each voltage and temperature corner fixed while threaded Python maintained uninterrupted environmental control.
- Reduced power consumption by 15%** via systematic voltage/temperature regularization during characterization.

PROJECTS

Solar Express – DC Power Optimizer | *Power Electronics, SPICE, Control Loop Design, Python, MPPT*

- Architected a **high power 600W** panel-level DC-DC power optimizer around a **4-switch buck-boost** mechanism, spanning a 10-80V input / 5-80V output envelope at 15A for robust **solar panel integration** into an efficient and scalable system.
- Validated the power stage via automated NGSpice simulations, achieving **99.5% pass-through** and **99.2% boost** efficiency.
- Optimized component selection via parametric loss-budget sweeps and verified key specs like NEC 690.12 rapid shutdown.

Automated Robot Race-Car | *C++, Embedded Systems, Analog Circuit Design, PID*

- Obtained real-time orientation and location data with **85% accuracy**: built circuitry (self-made rotary encoder for wheel-spin position mapping), integrated sensors (IMU + Magnetometer) with an STM32, applied RC filtering to cut motor noise.
- Applied PID control to autonomously track a preset optimal path and developed multi-sensor recalibration routines to re-localize the car after **1.5-foot ramp jumps** and obstacle encounters, overall achieving **fastest lap time in a 16-team competition**.